

Abstract

Accurate population data underpins some of the most consequential decisions in public life, from infrastructure investment and electoral redistricting to epidemic response and poverty measurement. Yet the primary instrument for collecting this data, the decennial census, is expensive, infrequent, and structurally unable to track the rapid demographic shifts that economic shocks can produce. Satellite imagery offers a promising alternative. Global, continuous, and updated at regular cadences, it can in principle provide the high-frequency population estimates that censuses cannot. This thesis evaluates whether geospatial foundation models can predict population density from satellite imagery at fine spatial resolutions, using Mexico as a test case. It assesses two state-of-the-art geospatial foundation models—Google’s AlphaEarth and the Clay model—on data from Mexico’s 2020 and 2010 censuses, covering over 100,000 urban AGEb and rural locality polygons. Using ridge regression and XGBoost prediction heads trained on model embeddings, this thesis asks two questions: can these models predict cross-sectional population density, and can they detect decadal population change? On cross-sectional prediction, AlphaEarth embeddings paired with XGBoost explain 73% of variation in log population density at the polygon level ($R^2 = 0.727$), rising up to 88% at the municipality level. These results exceed comparable benchmarks from task-specific models and other geospatial foundation models in the literature. Prediction quality varies systematically with polygon area, which is the strongest predictor of state-level accuracy ($r = 0.56$). On temporal prediction, decadal change proves substantially harder to model, with universally negative R^2 values. However, Pearson correlations of $r \approx 0.44$ – 0.53 indicate that both models reliably identify which areas are growing relative to which are shrinking. These findings position geospatial foundation models as cost-effective nowcasting tools capable of helping bridge the data gaps that rapid demographic change creates between census cycles.